Implement the Producer-Consumer problem using counting semaphores and POSIX threads. Evaluate synchronization correctness and system throughput under different buffer sizes.

Design a deadlock scenario involving multiple threads competing for shared resources. Demonstrate deadlock occurrence, identify the four necessary conditions for deadlock, and implement a deadlock prevention strategy using resource ordering.

```
Last login: Sat Sep 26 23:08:27 on ttys000
varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_Lab
varnikakomali@VARNIKAs-MacBook-Air OS_Lab % cd Practical
varnikakomali@VARNIKAs-MacBook-Air Practical % nano practical12.c
varnikakomali@VARNIKAs-MacBook-Air Practical % gcc practical12.c -o practical12 && ./practical12
=== Part 1: Producer-Consumer with Semaphores (Buffer Size: 5) ===

[Producer] Produced item 1 at index 0
[Consumer] Consumed item 1 from index 0
[Producer] Produced item 2 at index 1
[Consumer] Consumed item 2 from index 1
[Producer] Produced item 3 at index 2
[Consumer] Consumed item 3 from index 2
[Producer] Produced item 4 at index 3
[Producer] Produced item 5 at index 4
[Consumer] Consumed item 4 from index 3
[Producer] Produced item 6 at index 0
[Consumer] Consumed item 5 from index 4
[Producer] Produced item 7 at index 1
[Producer] Produced item 8 at index 2
[Consumer] Consumed item 6 from index 0
[Producer] Produced item 9 at index 3
[Consumer] Consumed item 7 from index 1
[Producer] Produced item 10 at index 4
[Consumer] Consumed item 8 from index 2
[Consumer] Consumed item 9 from index 3
[Consumer] Consumed item 10 from index 4

Producer-Consumer execution completed successfully without overflow or underflow.

=== Part 2: Deadlock Prevention via Resource Hierarchy Ordering ===

Deadlock requires 4 conditions: Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait.
Strategy: Enforce strict resource acquisition hierarchy (Res A -> Res B) to break Circular Wait.

[Thread 1] Acquired Resource A
[Thread 1] Acquired Resource B
[Thread 1] Released both resources
[Thread 2] Acquired Resource A (Resource Ordering enforced)
[Thread 2] Acquired Resource B
[Thread 2] Released both resources

Deadlock prevented successfully! Both threads acquired resources without circular deadlock.
varnikakomali@VARNIKAs-MacBook-Air Practical %
```